

Firoz Khan Patan

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TECHNICAL SKILLS

GenAI & LLMs	OpenAI API, Anthropic API, Prompt Engineering, LLM Evaluation, Puter.js
ML / AI	PyTorch, scikit-learn, Embeddings, SVM, Neural Networks (Backprop), K-Means, Recommender Systems, Cross-Validation
Frameworks	FastAPI, React, Next.js, Node.js, TensorFlow, Keras, NumPy, Pandas, Matplotlib
Prog/DevOps	Python, JavaScript, Java, C AWS, Docker, GitHub Actions CI/CD, REST APIs, Linux, Git, Google Colab

EXPERIENCE

Software Developer Intern

Jan 2024 – Jun 2024

Prochal Technologies Pvt. Ltd., Hyderabad, India

- Collaborated with founding engineers to build features for a scalable assessment platform; designed **RESTful APIs** and role-based dashboards improving data flow and placement tracking efficiency.
- Developed a template-driven resume builder in **Next.js**, improving user experience and reducing manual effort for students and recruiters.

PROJECTS

AI News Aggregator — GenAI Backend Application

Apr 2026 – Present

- Built a production-grade AI pipeline that scrapes AI news from **YouTube channels and blogs** via RSS/Atom feeds and stores normalized content in **PostgreSQL** (sources, posts, user_profiles tables).
- Integrated **OpenAI GPT-4o** to generate personalized daily digests tailored to each user's interest profile; automated delivery via **SMTP email** with HTML-formatted digest output.
- Designed and exposed a **REST API** (FastAPI) for source/user management and manual pipeline runs; fully containerized with **Docker** and orchestrated via **Docker Compose**.

Resume Analyzer AI — Web Application

Dec 2025 – Feb 2026

- Built an AI-powered resume analysis app with **LLM**-driven evaluation, ATS-style scoring, and job-fit prediction using **React**, **Tailwind CSS**, and **Puter.js**; integrated cloud authentication, file storage, and PDF-to-image conversion.
- Designed a responsive multi-page interface presenting resume scores across tone, style, content, structure, skills, and job-fit categories tailored to specific job descriptions.

TimeMCL Learning Implementation — ML Research Paper Implementation

2025

- Implemented the core method from "*Winner-Takes-All for Multivariate Probabilistic Time Series Forecasting*" (Cortes et al., ICML 2025), a probabilistic forecasting approach that predicts multiple plausible futures in a single forward pass using **Multiple Choice Learning**.
- Built a shared **LSTM** encoder with multiple forecast heads and optional score heads in **PyTorch**; implemented **WTA**, relaxed WTA, and annealed softmin loss variants where each training target updates only the closest prediction head.
- Designed synthetic multimodal time-series data generation for rapid experimentation; produced SVG forecast visualizations comparing predicted trajectory distributions against held-out context windows.

LetsChat — TCP Chat System

Apr 2025

- Engineered a multi-user real-time chat system over **TCP** using **select()/poll()** for multiplexed I/O supporting concurrent private and group chats; containerized with **Docker** for automated deployment.

EDUCATION

University of Georgia, Athens, GA

Aug 2024 – May 2026

Masters, Computer Science

GPA: 3.52/4.00

Sathyabama Institute of Science and Technology, Chennai, India

Aug 2020 – Apr 2024

Bachelor of Engineering, Computer Science

CGPA: 8.26/10.00

PUBLICATIONS

Firoz Khan Patan, S. L. JanyShabu, V. Peddineni. "Crime Analysis and Detection Based on Location Using Machine Learning." *AIP Conference Proceedings*, Vol. 3257, Art. 020033, July 2025. Presented at ICCCAI 2024.